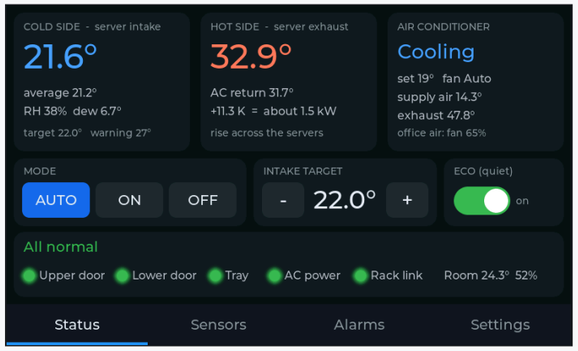
CP-SRA16-ELC-001 - System overview

UPPER DOOR - outside the steel, so Wi-Fi works

TOUCHSCREEN Waveshare ESP32-S3-Touch-LCD-4.3B

4.3 in 800 x 480 IPS, capacitive touch, in the printed pod P6
Shows cold / hot side, AC state, every sensor, alarms
Sets mode (auto / on / off), intake target, eco, fan, limits
Modbus client on RS485: polls the node every 2 s
Wi-Fi: Home Assistant API, web page, over-the-air updates
Room SHT41 in the pod vent; upper door contact on DI0
Buzzer on DO0; alarm output DO1 for a beacon
Real-time clock keeps the time through power cuts



status screen - firmware preview, real size 95 x 57 mm

HOME ASSISTANT (optional, over Wi-Fi)

history, phone alerts, automations; the 'IT shutdown request'
can run NUT / ssh shutdowns of the servers (software path)

Pod P6: 128 x 91 x 36 mm, PETG, on two M4 rivnuts in the upper door;
the cable enters through the door's 20 mm grommet (DXF DU1).

W1 door bus

+12 V
0 V
A
B
GX16-4

4-core shielded, 1.5 m
RS485 Modbus RTU
19 200 baud 8N1
through a hinge loop;
the GX16 plug lets
the door lift off

INSIDE THE RACK

RACK NODE ESP32-S3-DevKitC-1 on a carrier, box P7

Runs the cooling on its own: needs neither the screen nor
the network to keep the rack cool. Modbus server 1.
Reads 6 DS18B20 probes + an SHT41 every 10 s
Decides every 5 s: AC on / off, setpoint, fan, eco
Sends the AC's own remote frames by IR (coolix / midea)
Alarms; closes the SHUTDOWN relay K1 when critical
12 V in, 5 V buck, 3.3 V on the DevKit; about 1 W
Fail-safe: anything unknown means AC on

pins, schematic and board: sheet 2 - where it all goes: sheet 4

AIR CONDITIONER Dimplex GDC14RBA

keeps its own thermostat and compressor timer: the node
sends only what its remote would. The stick-on emitter sits
on its IR window; the AC still rolls out (unplug the lead)

POWER 12 V 2 A plug pack on the UPS-backed PDU

node + screen about 3.8 W typical; everything is SELV
on the UPS, so the node can still ask for a clean shutdown
the AC stays on its own wall socket

WHEN SOMETHING FAILS

screen or Wi-Fi down: the node keeps cooling, unchanged
node down: the screen shows 'Rack link' red and beeps
probe lost: alarm, and the AC is kept on
too hot 120 s: K1 + Home Assistant ask for an IT shutdown

J5 **T1 T2 cold side top / middle**
front plenum: server intake

J5 **T3 AC supply air**
hood outlet

J4 **SHT41 cold-side humidity**
dew point: condensation watch

J6 **T4 hot side top**
rear plenum: server exhaust

J6 **T5 AC return air**
rear plenum, above the return opening

J6 **T6 AC exhaust**
condenser elbow: is the compressor on?

J7 **IR emitter -> AC receiver**
on / off, setpoint, fan

J8 **Tray float switch**
water in the drip tray

J9 **Lower door contact**
door left open

J10 **Mains detect (option)**
AC socket live: faster restart

J3 **SHUTDOWN contact K1**
to the UPS / server signal input

Why ESP32 and not a Raspberry Pi 5

	ESP32-S3 kit	Pi 5 instead
Boot after a power cut	about 1 s, nothing to corrupt	20-40 s, SD card at risk
Heat inside a sealed rack	about 1 W	5-10 W plus a fan
IR timing	hardware (RMT peripheral)	needs a driver or helper chip
12 V from the UPS	direct (buck / 7-36 V panel)	5 V 5 A supply
Touchscreen	panel with RS485 + isolated I/O	DSI panel + USB adapters
Home Assistant	native ESPHome API	MQTT / custom code
Parts cost	about AUD 358 for the kit	about AUD 250 more

A Pi 5 is still a fine Home Assistant host - outside the rack, on the network.

What is where

firmware/sra16-node.yaml	rack node (ESPHome 2026.6)
firmware/sra16-hmi.yaml, hmi/ui.yaml	touchscreen and its screens
firmware/components/sra_control/	control core (C++, header only)
firmware/test/	unit tests, simulation report
tools/sim_controller.py	thermal model, drives the real core
cad/print P6 P7 P8, P1-1 port	pod, node box, clips, hood port
bom/electronics*.csv	kit parts, harness
docs/ELECTRONICS.md	build, install, commissioning

Key figures

Probes	6 x DS18B20 (0.5 K) + 2 x SHT41
Decision cycle	5 s; IR at most every 10 s
Bus	RS485, 1.5 m, 19 200 baud
Supply	12 V, 3.8 W typical
Simulation	26/26 checks pass
Unit tests	156 checks, all pass
Firmware size	node 48 % flash, screen 17 %

Typical sources (Australia)

Panel, DevKit, SHT41, buck	Core Electronics, DigiKey, Mouser
Probes, float, reeds, GX16, IR	Altronics, Jaycar, AliExpress
Terminals, board, R / C, relay	Altronics, Element14, DigiKey
Plug pack (RCM approved)	Jaycar, Altronics

Kit cost (bom/electronics.csv, indicative AUD)

Touchscreen	98
Rack node	74
Sensors	61
AC control	12
Cables	40
Power	33
Office air	41
Kit total	358
Options: mains detect, beacon	71

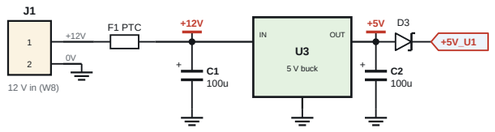
One decision, every 5 s

1 read the probes (every 10 s); 10-minute averages smooth the
compressor's own cycling
2 choose AC on / off, setpoint and fan (sheet 5)
3 send by IR only if something changed (at most every 10 s)
4 check it worked: supply air below return, exhaust above room
5 not cooling: re-send 3 times, then alarm; critical: ask for
the IT to shut down (K1 + Home Assistant)

PROJECT	SRA-16 Silent AC Rack - control electronics kit
TITLE	System overview, why ESP32, files
DOC NO.	CP-SRA16-ELC-001 SHEET 1 OF 5 REV B
SCALE	NTS @ A3 UNITS mm LAYOUT v1.3.0
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	PROTOTYPE - SELV only (12 V); no mains wiring in the kit

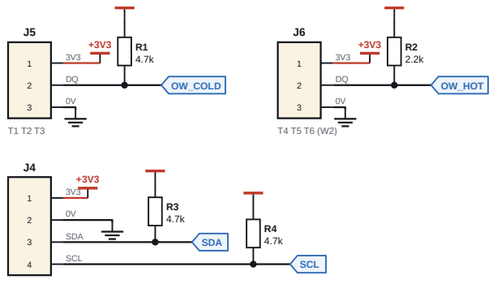
CP-SRA16-ELC-001 - Rack node carrier

POWER

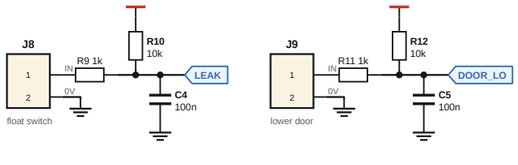


+12V also feeds J2 (door bus, after F1) and K1. U1's own LDO makes +3V3.

PROBES (1-Wire) + HUMIDITY (I2C)

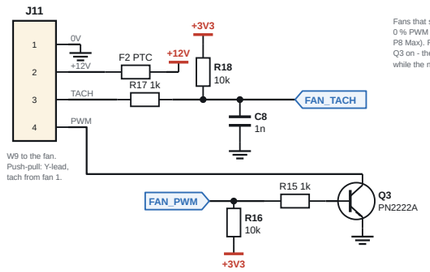


INPUTS (contact to 0 V)



Polarity is set in the firmware; rails without a name are +3V3.

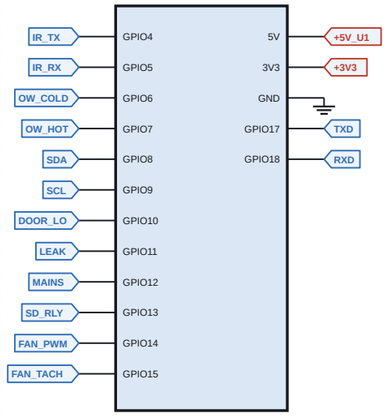
OFFICE AIR FAN (J11, 4-pin PC fan)



Fans that stop at 0 % PWM (NF-A8 PWM, P8 Max). R16 keeps Q3 on - the fan off - while the node boots.

U1 ESP32-S3-DevKitC-1

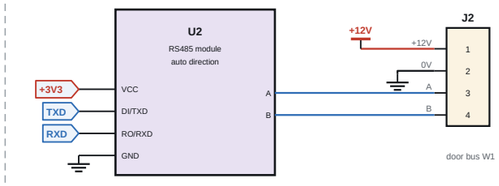
N8R8: GPIO35-37 are the PSRAM's



3.3 V logic: never 5 V on a GPIO

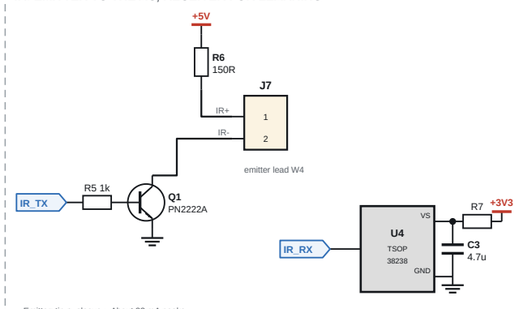
2 x 22-way sockets
USB-C ports up (lid side)
for the first flash

RS485 TO THE TOUCHSCREEN



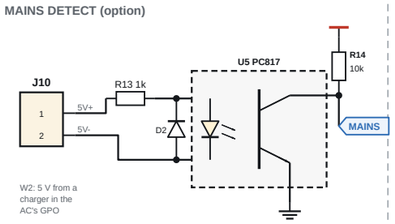
C7 100n at U2's VCC. 120R at each end (module + panel).

IR: EMITTER TO THE AC, RECEIVER FOR LEARNING



Emitter: tip +, sleeve -. About 22 mA peaks.

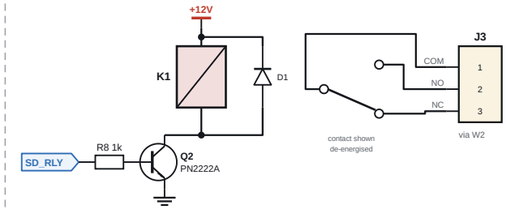
MAINS DETECT (option)



W2: 5 V from a charger in the AC's GPO

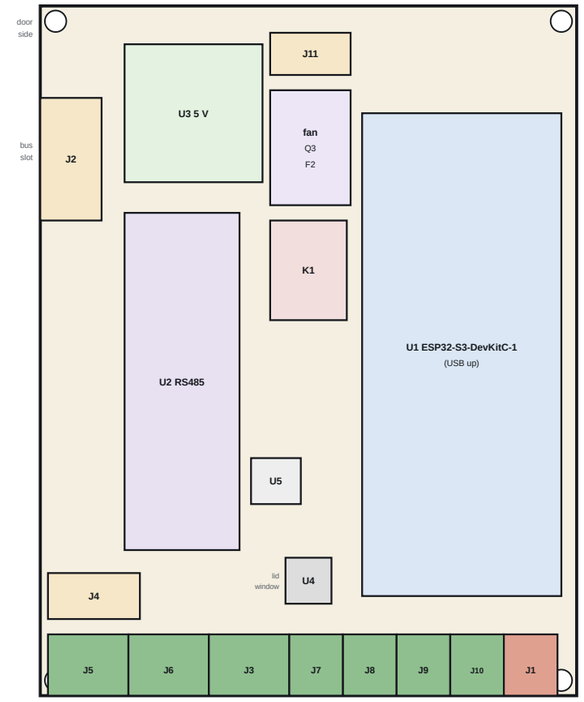
Isolated: the charger only lights U5's LED. C6 100n from MAINS to 0 V. Firmware: mains_fitted: "true" once wired.

SHUTDOWN RELAY (dry contact)



COM-NO closes while the node asks for the IT to shut down (latched until 'Reset alarms'). For UPS / server signal inputs only: 1 A, 30 V DC.

BOARD 70 x 90, component side, 1.25 : 1



Height in P7 (26.5 inside): standoff 5 board 1.6 sockets 8.5 DevKit 1.6 module 3.2 = 19.9 + base 2.5 = 22.4 K1 10, U3 3

terminal row: wires enter from below (box slots at x -22, 0, +22)

upright in box P7, magnets at the right-hand end

Ref	Part	Use
U1	ESP32-S3-DevKitC-1-N8R8	the controller, on 2 x 22-way sockets
U2	TTL-RS485 module, auto direction, 3.3 V	SP3485/MAX3485 class, A/B to J2
U3	5 V 1 A step-down, 7-36 V in	Pololu D24V10F5 or equal
U4	TSOP38238	IR receiver 38 kHz, under the lid window
U5	PC817	optocoupler, mains detect
K1	Omron G5V-1-DC12	SHUTDOWN relay, SPDT 1 A 30 VDC
Q1, Q2	PN2222A	IR emitter and relay drivers
D1	1N4148	K1 flyback
D2	1N4148	U5 LED reverse guard
D3	1N5819	5 V into U1: blocks back-feed from the USB port
F1	PTC 0.5 A hold	door bus 12 V
R1	4.7k	1-Wire cold pull-up
R2	2.2k	1-Wire hot pull-up (longer bus)
R3, R4	4.7k	I2C pull-ups
R5, R8	1k	Q1, Q2 base
R6	150R 0.25 W	IR emitter current, about 22 mA peak
R7	100R	U4 supply filter
R9, R11, R13	1k	input series / U5 LED
R10, R12, R14	10k	input pull-ups
C1	100uF 25 V	12 V bulk
C2	100uF 10 V	5 V bulk
C3	4.7uF	U4 supply
C4-C7	100nF	input filters, U2 decoupling
Q3	PN2222A	office fan PWM, open collector (J11-4)
F2	PTC 0.5 A hold	office fan 12 V (J11-2)
R15, R17	1k	Q3 base; tach series
R16, R18	10k	FAN_PWM pull-up (fan held off while the node boots); tach pull-up
C8	1nF	tach filter

Pin	Net	Goes to	Function
GPIO4	IR_TX	Q1 base (R5)	IR emitter, 38 kHz carrier (remote_transmitter)
GPIO5	IR_RX	U4 OUT	IR receiver: learning, protocol check (remote_receiver)
GPIO6	OW_COLD	J5-2	1-Wire, cold probes T1-T3 (R1 4.7k pull-up)
GPIO7	OW_HOT	J6-2	1-Wire, hot probes T4-T6 (R2 2.2k pull-up)
GPIO8	SDA	J4-3	I2C data, SHT41 cold-side humidity
GPIO9	SCL	J4-4	I2C clock (50 kHz on the 0.5 m lead)
GPIO10	DOOR_LO	J9-1 (R11)	lower door contact, closed when shut
GPIO11	LEAK	J8-1 (R9)	tray float switch, closes on water
GPIO12	MAINS	U5 collector	AC socket live: optocoupler pulls low
GPIO13	SD_RLY	Q2 base (R8)	SHUTDOWN relay K1 -> J3
GPIO14	FAN_PWM	Q3 base (R15; R16 holds it on)	office fan PWM 25 kHz, open collector to J11-4
GPIO15	FAN_TACH	J11-3 (R17; R18 10k to 3V3)	office fan speed, 2 pulses per turn
GPIO17	TXD	U2 DI/TXD	RS485 transmit (UART1, 19200 8N1)
GPIO18	RXD	U2 RO/RXD	RS485 receive

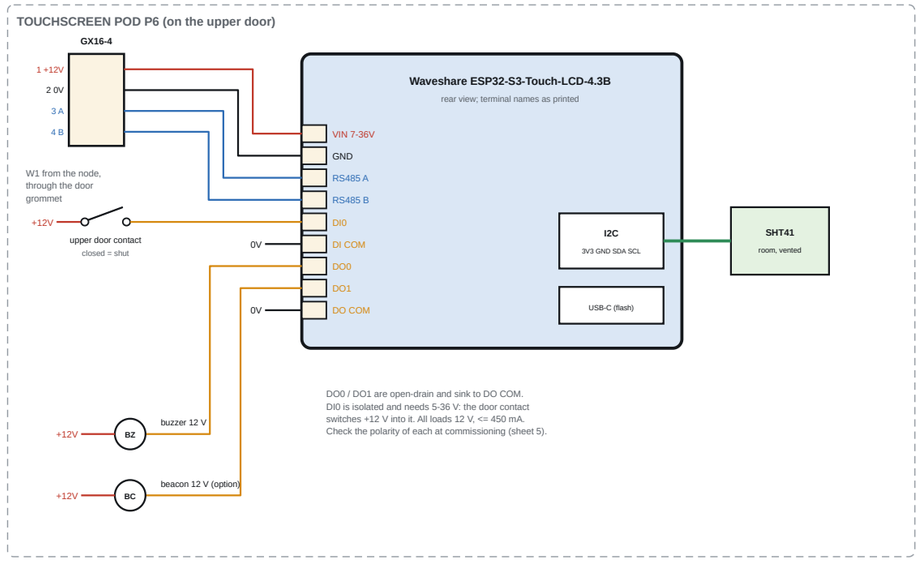
Conn	Type	Pins 1..n	To
J1	2-way terminal 3.5 mm	+12V / 0V	12 V in: plug pack on the UPS-backed PDU (W8)
J2	4-way terminal 3.5 mm	+12V / 0V / A / B	door bus W1 to the touchscreen
J3	3-way terminal 3.5 mm	COM / NO / NC	SHUTDOWN dry contact (K1) on W2
J4	JST-XH 4-way	3V3 / 0V / SDA / SCL	SHT41 cold-side humidity (W7)
J5	3-way terminal 3.5 mm	3V3 / DQ / 0V	cold probes T1 T2 T3 (W3)
J6	3-way terminal 3.5 mm	3V3 / DQ / 0V	hot probes T4 T5 T6 on harness W2
J7	2-way terminal 3.5 mm	IR+ / IR-	IR emitter socket lead W4
J8	2-way terminal 3.5 mm	LEAK / 0V	tray float switch (W5)
J9	2-way terminal 3.5 mm	DOOR / 0V	lower door contact (W6)
J10	2-way terminal 3.5 mm	5V+ / 5V-	mains detect, optional (W2)
J11	4-pin PC fan header, 2.54 mm	0V / +12V / TACH / PWM	office air fan (W9); 2nd fan on a Y-lead

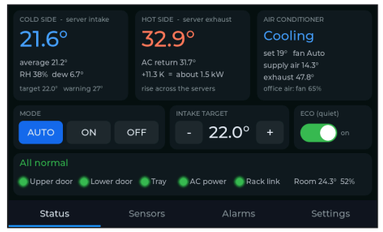
Build: sockets for U1, then U3 and U2, then the terminal row. With U1 out, power J1: +5 V, +12 V on J2; fit U1: +3V3.

Keep the 12 V wiring on the board's door-side half; 1-Wire and I2C short and away from K1. Pull-ups and filters as drawn.

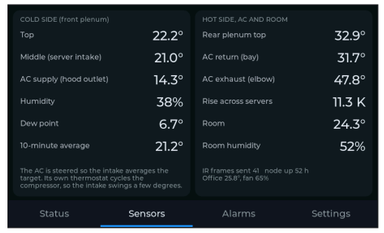
PROJECT	SRA-16 Silent AC Rack - control electronics kit
TITLE	Node carrier: schematic, board, parts
DOC NO.	CP-SRA16-ELC-001 SHEET 2 OF 5 REV B
SCALE	NTS @ A3 UNITS mm LAYOUT v1.3.0
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	PROTOTYPE - SELV only (12 V); no mains wiring in the kit

CP-SRA16-ELC-001 - Touchscreen pod, harness, Modbus map

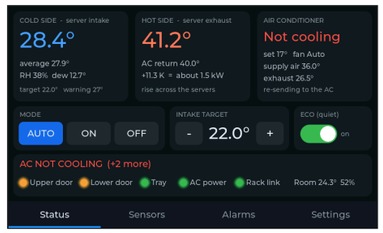




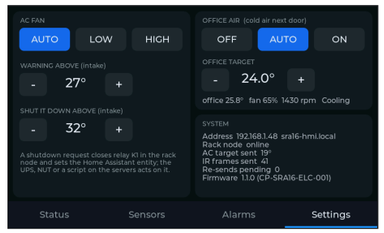
Status screen



Sensors screen



Alarm screen



Settings screen

W1 door bus, GX16-4

Pin	Signal	Core
1	+12V	red
2	0V	black
3	A	white / green
4	B	green / white

Shield: 0 V at the node end only. The plug sits on the cabinet side of the hinge loop: unplug it before lifting the door off.

Power budget

Load	Rail	mA	peak
Touchscreen panel (backlight on)	12 V	210	260
ESP32-S3 node incl. Wi-Fi	3.3 V (via 5 V)	90	350
Probes, SHT41, RS485, IR receiver	3.3 V	12	20
IR emitter (while sending)	5 V	0	22
Relay K1 (shutdown only)	12 V	0	13
Buzzer / beacon (alarm only)	12 V	0	30
Office fan, NF-A8 PWM (2 x P8 Max: about 0.6 A)	12 V	60	80

peak 0.58 A at 12 V (plug pack 2 A), 5 V rail 392 mA peak (U3 1 A);
typical 3.8 W.
Plug pack 12 V 2 A, regulated, RCM approved.

HARNESS SCHEDULE (bom/electronics_harness.csv; routes on sheet 4)

W	From	To	Cable	m	Route
W1	J2 (node)	touchscreen terminals	4-core 0.22 shielded, GX16-4 inline	1.5	node box front slot - front-left corner of the front plenum - hinge loop 300 mm in spiral wrap at z 1450 (GX16 on the cabinet side) - upper door
					grommet - pod
W2	J6, J3, J10 (node)	T4, T5, T6; UPS contact; mains charger	8-core 0.22 shielded	3.5	bottom slot - down the front-left corner - hood port - back along the bay's left wall under the shelf (T6 on a branch tied along the exhaust duct) -
					up through the shelf's return opening - rear-left rail spacer (T5, then T4) - cable entry
W3	J5 (node)	T1, T2, T3	probe leads, 1 m	1.0	front-left rail spacer face, clipped
W4	J7 (node)	IR emitter (plugs in)	3.5 mm socket lead	1.5	hood port - socket tied in the bay's top-front-left corner - emitter over the AC's IR receiver
W5	J8 (node)	tray float switch	2-core 0.22	1.8	hood port - down the AC's left side gap - tray, front left
W6	J9 (node)	lower door contact	2-core 0.22	1.0	hood port - under the hood's front edge
W7	J4 (node)	SHT41 cold side	4-core 0.22	0.5	up the front-left rail spacer, P8-1 clip at z 1300
W8	J1 (node)	12 V plug pack	DC lead + 2 m extension	3.5	PDU - (rear: down the return opening, bay, hood port) - bottom slot
W9	J11 (node)	office supply fan (P9, front plenum, right side)	4-pin PWM fan extension	1.0	node box bottom slot - along the shelf's front edge to the right rail spacer - up its front face to z 1290 - into the supply sleeve through the notch
					at the bottom of its grille
W10	T7 office probe	J6 via W2, spliced at T5	probe lead, 3 m	3.0	office: return wall grille P12 - inside the return duct - return sleeve slot (rear plenum, right) - along the rear plenum floor to the rear-left rail
					spacer - spliced into W2 with T4 and T5
W11	Y-lead on W9 (option)	second fan, return sleeve	4-core 0.22	3.0	with W2: hood port - bay - the shelf's return opening - rear plenum floor - return sleeve (push-pull option)

MODBUS MAP node = server 1, holding registers, 19 200 8N1; -32768 = no reading

Reg	Name	Type	Scale	R/W
0x00	Cold side (mean T1, T2)	S16	0.1 degC	R
0x01	Hot side (max T4, T5)	S16	0.1 degC	R
0x02	T1 cold top	S16	0.1 degC	R
0x03	T2 cold middle	S16	0.1 degC	R
0x04	T3 AC supply	S16	0.1 degC	R
0x05	T4 hot top	S16	0.1 degC	R
0x06	T5 AC return	S16	0.1 degC	R
0x07	T6 AC exhaust	S16	0.1 degC	R
0x08	Cold side RH	S16	0.1 %	R
0x09	Dew point	S16	0.1 degC	R
0x0A	Cold side average	S16	0.1 degC	R
0x0B	Rise across the servers	S16	0.1 K	R
0x0C	Setpoint sent	S16	0.1 degC	R
0x0D	State (0-8)	U16	enum	R
0x0E	Alarm bits	U16	bits 0-12	R
0x0F	Flags	U16	bits 0-8	R
0x10	Retries	U16	count	R
0x11	Uptime	U16	min	R
0x12	IR frames sent	U16	count	R
0x13	Firmware version	U16	110 = 1.1.0	R

Flags 0x0F: b0 AC on (commanded), b1 eco, b2 shutdown request, b3 lower door open, b4 water,
b5 AC socket live, b6 compressor running, b7 compressor state known, b8 mains detect fitted.

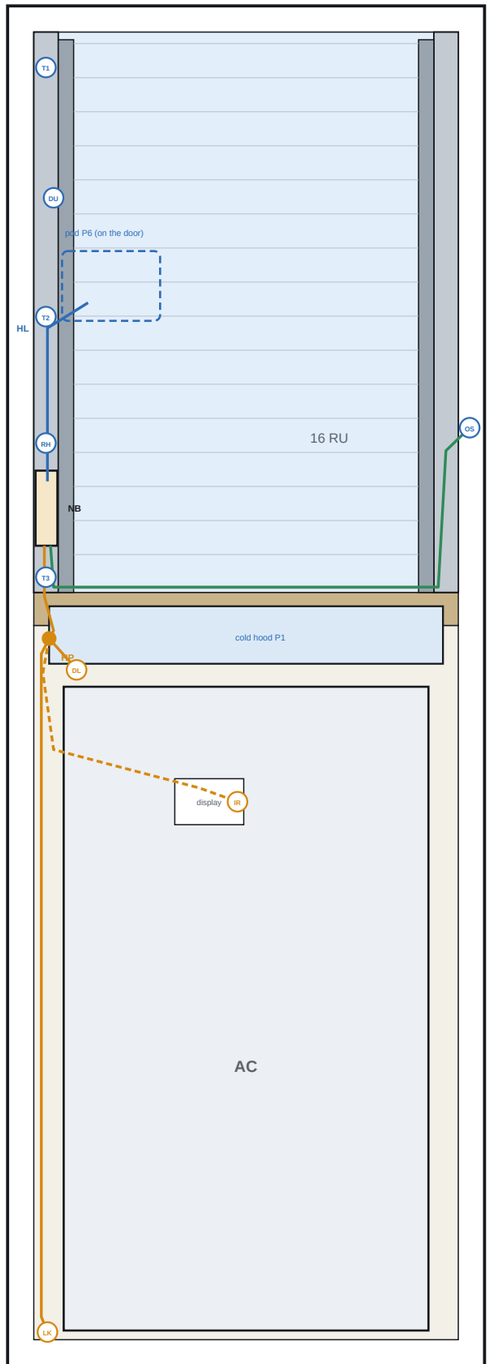
Reg	Name	Type	Scale	R/W
0x14	Office (T7)	S16	0.1 degC	R
0x15	Office fan	U16	%	R
0x16	Office fan speed	U16	rpm	R
0x17	Office air state (0-5)	U16	enum	R
0x18	Cold-side target in use	S16	0.1 degC	R
0x20	Mode: 0 auto, 1 on, 2 off	U16	enum	R/W
0x21	Intake target	S16	0.1 degC, 18-27	R/W
0x22	Eco allowed	U16	0/1	R/W
0x23	AC fan: 0 auto, 1 low, 2 high	U16	enum	R/W
0x24	Intake warning	S16	0.1 degC, 24-35	R/W
0x25	Shutdown limit	S16	0.1 degC, 28-40	R/W
0x26	Office air: 0 off, 1 auto, 2 on	U16	enum	R/W
0x27	Office target	S16	0.1 degC, 20-28	R/W
0x28	Office supply (cold side while it calls)	S16	0.1 degC, 16-22	R/W
0x29	Office fan maximum	U16	%, 30-100	R/W
0x30	Upper door open (from the screen)	U16	0/1	W
0x31	Room temperature	S16	0.1 degC	W
0x32	Command: 1 send IR now, 2 reset alarms	U16		W
0x33	Heartbeat	U16	count	W
0x34	Room humidity (from the screen)	S16	0.1 %	W

HOME ASSISTANT (from the screen, sra16-hmi)

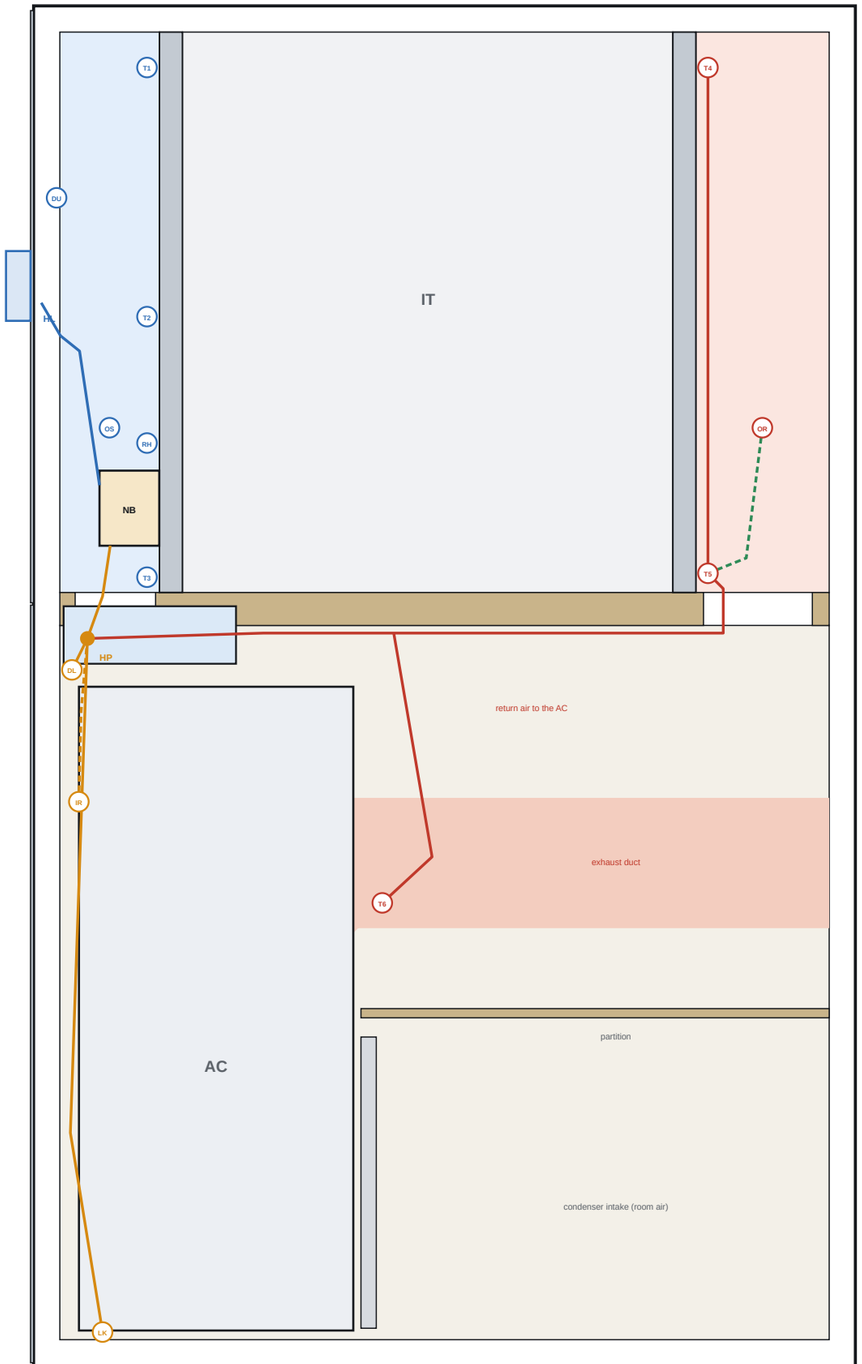
sensor	cold / hot side, T1-T6, room, RH, dew point, cold average,
	rise across the servers, setpoint sent, alarm bits, IR frames,
	uptime; office T7, office fan % and rpm
binary_sensor	9 alarms (intake warning / critical, not cooling, water, door,
	probe, condensation, AC socket, office fan), upper door, rack
	link, IT shutdown request
select	cooling mode (auto / on / off), AC fan (auto / low / high),
	office air (off / auto / on)
number	intake target, intake warning, shutdown limit; office target,
	office supply, office fan max
switch	eco (quiet at low load)
button	re-send state to the AC, reset alarms
text_sensor	cooling state, office air state, IP address

PROJECT	SRA-16 Silent AC Rack - control electronics kit
TITLE	Touchscreen pod wiring, harness, Modbus map
DOC NO.	CP-SRA16-ELC-001 SHEET 3 OF 5 REV B
SCALE	NTS @ A3 UNITS mm LAYOUT v1.3.0
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	PROTOTYPE - SELV only (12 V); no mains wiring in the kit

CP-SRA16-ELC-001 - Placement in the rack, cable routes



FRONT, doors off 1:8



LEFT SECTION, looking at the left wall 1:8

WHERE EVERYTHING GOES

Id	What	x / y / z	How
T1	Cold side top	50 / 148 / 1790	P8-1 clip on the front-left rail spacer; probe across the plenum
T2	Cold side middle	50 / 148 / 1465	P8-1 clip, same face
T3	AC supply	50 / 148 / 1125	P8-1 clip just above the shelf, tip over the hood outlet
RH	SHT41 cold side	50 / 148 / 1300	lead in a P8-1 clip, board hanging free in the air
T4	Hot side top	50 / 880 / 1790	P8-1 clip on the rear-left rail spacer, rear face
T5	AC return	50 / 880 / 1130	P8-1 clip on the rear-left rail spacer, just above the return opening
T6	AC exhaust	442 / 455 / 700	6.5 mm hole in the elbow's outer bend, probe 25 mm in, silicone
LK	Tray float switch	52 / 90 / 140	on a small alu angle bonded to the tray, front-left corner
DL	Lower door contact	90 / 50 / 1004	switch under the hood's front edge, magnet on the door lining
DU	Upper door contact	60 / 30 / 1620	switch on the door lining at the hinge side, magnet on the side lining
IR	IR emitter	300 / 59 / 832	stuck on the AC front over its IR receiver (by the display)
NB	Node box (P7)	51 / 125 / 1215	magnets on the front-left rail spacer, front face
POD	Touchscreen (P6)	135 / -30 / 1505	outside the upper door, 2 x M4 rivnuts
HP	Hood port	54 / 70 / 1045	P1-1 left end, 25 mm membrane grommet
HL	Hinge loop + GX16	40 / 50 / 1450	300 mm loop in spiral wrap, between the hinges
OS	Office supply port	603 / 99 / 1320	sleeve P9 in the right lining, fan blowing out; W9 from J11
OR	Office return port	603 / 951 / 1320	office air back in; probe T7's lead (W10) comes in with it

x from the left skin, y back from the front, z up from the floor (rack_layout v1.3.0).
Cold side (blue): front plenum. Hot side (red): rear plenum and the bay above the partition, which carries the return air down to the AC. Everything that crosses from the front plenum to the bay goes through the hood port HP; no other holes are needed. Keep probe leads 50 mm from mains cords, tied but not tensioned, and out of the condenser intake.

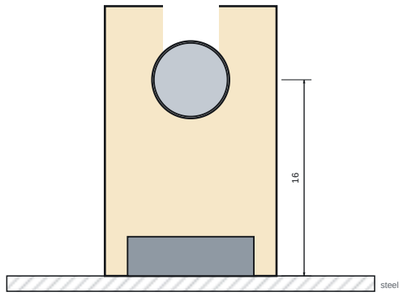
- W1 door bus; cold-side leads
- W4 IR, W5 leak, W6 lower door, via HP
- W2 rear harness: T6, T5, T4, UPS, mains
- office air: W9 fan (front view), W10 probe T7 (dashed, far wall)

A HOOD PORT 1:4



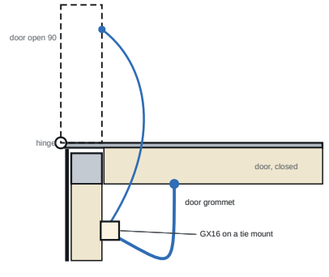
D25 through, wall thinned to 2 mm over D35 inside; 25 mm membrane grommet - pierce one hole per cable. Front face on the left.

C PROBE CLIP P8-1 2:1



6 mm probe snaps in from the top; 10 x 3 magnet glued in the base.

B HINGE LOOP, PLAN AT z 1450 1:6

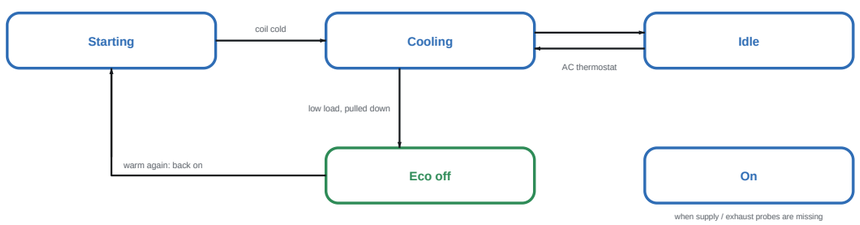


300 mm of W1 in 10 mm spiral wrap runs from the door grommet to a tie mount on the side lining, inside the seal line and half way between the hinges. Open the door fully before tying off: no tension when open, nothing pinched when shut. Thin line: the cable with the door open.

PROJECT	SRA-16 Silent AC Rack - control electronics kit
TITLE	Placement in the rack, cable routes
DOC NO.	CP-SRA16-ELC-001 SHEET 4 OF 5 REV B
SCALE	1:8 views, details as noted @ A3 UNITS mm LAYOUT v1.3.0
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	PROTOTYPE - SELV only (12 V); no mains wiring in the kit

CP-SRA16-ELC-001 - Control logic, alarms, commissioning

STATES (rack node, decided every 5 s)





Each returns through Starting once its cause has cleared; the AC's 3-minute restart delay is respected.

State	Meaning
Starting	just switched on, the coil is still cooling down
Cooling	compressor running
Idle	AC on, its thermostat satisfied
On	AC on, compressor state unknown (no supply / exhaust probe)
Eco off	switched off at low load, back on when it warms
Off	switched off by the user
Not cooling	should cool, does not after 3 re-sends
No AC power	mains-detect input dead
CRITICAL	intake above the shutdown limit

Normal load: AC on; its setpoint (whole degrees) is trimmed so that the 10-minute average intake meets the target - the AC's own thermostat cycles the compressor. Low load (the servers warm the air by less than 4 K): eco pulls the rack down to target - 2, switches the AC off and back on at target + 2 or a hot side of 32. Unknown readings or over-temperature always mean AC on.

ALARMS (register 0x0E: bit)

Bit	Alarm	When	Action
0	intake warning	cold side >= warning (27)	screen, beep
1	intake critical	cold side >= limit (32)	beep; shutdown after 120 s
2	hot side warning	hot side >= 45	screen
3	AC not cooling	3 re-sends, still not cooling	beep
4	water in tray	float switch closed 5 s	beep
5	door left open	a door open 10 min	screen
6	probe missing	no reading on the cold or the hot side	screen; AC forced on
7	condensation risk	dew point 1 K over supply air for 10 min	screen
8	user OFF overridden	off, but intake hit the warning	screen
9	IT shutdown	critical for 120 s (latched)	K1 closes, beep
10	AC socket dead	mains-detect input low	beep
11	link	no screen heartbeat 60 s / node not answering	screen
12	office fan	told to run, under 300 rpm for 60 s	screen

OFFICE AIR (register 0x17; docs/OFFICE_AIR.md)

State	Meaning
Off	office air switched off (or no office kit)
Idle	auto: the office is at its target
Cooling	the fan runs: cold-side air to the office, its air back to the hot side
Waiting	the office calls; the cold side is not yet 1.5 K colder than it (or the fan rests)
Rack first	the office calls, but the rack is warm (cold side 1.5 K over target), an alarm is up, or the AC is off by the user / without power
No office probe	auto, but no reading from T7

The office calls when T7 passes its target + 0.5 K. While it calls and the rack is in hand, eco is held off and the cold side is pulled down to the office supply temperature (18, or 2 K over the room dew point); the fan runs, faster the warmer the office, while the cold side is 1.5 K colder than the office. A warm rack, an alarm or the user's OFF stops it at once: the rack always comes first.

SETTINGS (defaults; screen or Home Assistant)

Setting	Default	Notes
Intake target	22.0 degC	average of the cold side the AC is trimmed to
Warning / shutdown limit	27 / 32 degC	ASHRAE A1 recommended max / hard limit
Eco	on	AC off at low load: rise across the servers < 4 K
Eco band	target -2 / +2 K	off when pulled down, on again when warm (or hot side 32)
Min on / off	15 / 5 min	compressor-friendly cycling
Setpoint range	17-30 degC	first guess target + 3, trimmed in whole degrees
Trim	20 min (2 min when 3 K hot)	on the mean error over the window
Not-cooling check	15 min (3 min when hot)	then re-send, 3 times, then alarm
Shutdown delay	120 s	cold side above the limit before K1 closes
IR send gap	>= 10 s	never floods the AC
Office air	Auto (kit fitted)	Off / Auto / On (On ignores the office probe)
Office target	24 degC	calls above 24.5, satisfied below 23.5
Office supply	18 degC	cold-side target while it calls; >= room dew point + 2 K
Office fan	30-100 %	over 2 K of call; 1.5 K colder than the office; 2 min run

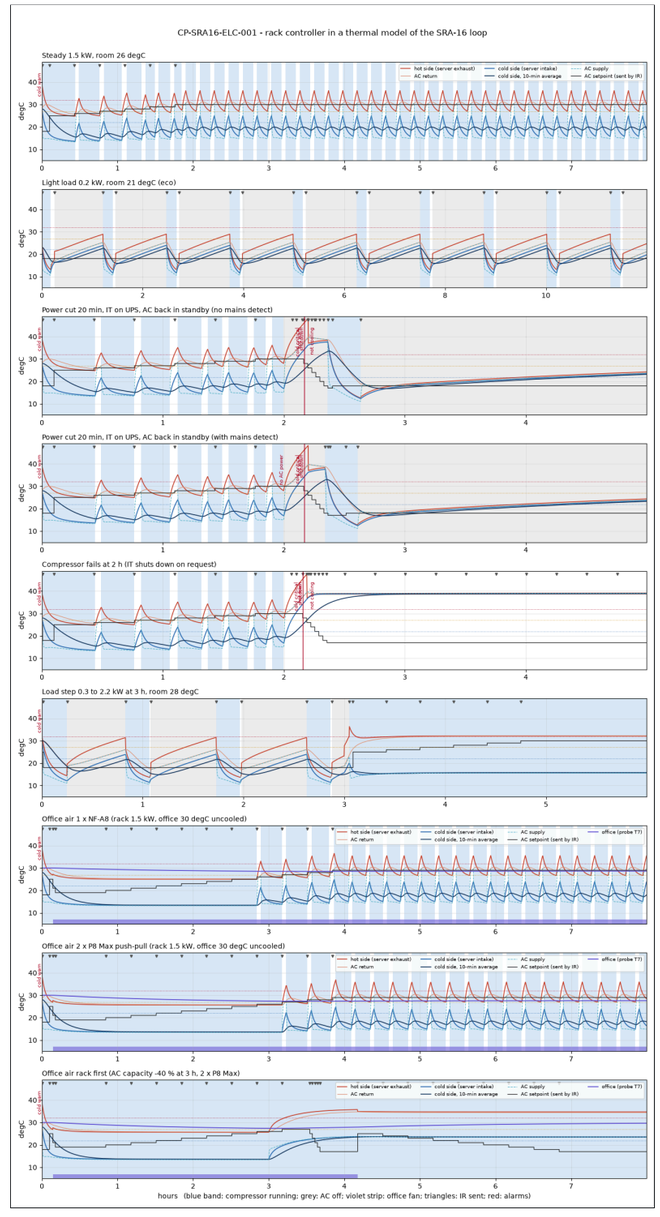
SIMULATION (the real control core in a thermal model)

Scenario	Check		Result
Steady 1.5 kW	cold side averages at or below the target	OK	average 19.4 degC (target 22), swing 15.8-25.0
	cold side peaks below the warning	OK	max 25.0 degC
	no alarms once settled	OK	none
	setpoint steady (no hunting)	OK	0 setpoint changes in hours 2-8
Light load 0.2 kW (eco)	AC off most of the time	OK	on 20% of hours 2-12
	AC cycles, never faster than its minimum times	OK	19 switches, shortest gap 15 min
	cold side never above the warning	OK	max 24.0 degC
Power cut 20 min (no mains detect)	AC restarted after the power returned	OK	1.8 min after power back (limit 20 min)
	IT shutdown requested while the AC was down	OK	10 min into the cut
	cold side peak held under 40 degC	OK	peak 37.6 degC
Power cut 20 min (with mains detect)	AC restarted after the power returned	OK	0.5 min after power back (limit 1 min)
	IT shutdown requested while the AC was down	OK	10 min into the cut
	cold side peak held under 40 degC	OK	peak 37.5 degC
Compressor fails at 2 h	not-cooling alarm within 20 min	OK	15 min after the failure
	shutdown requested	OK	9 min after the failure
	IT stopped before the cold side reached 38 degC	OK	cold side 36.5 degC when the IT stopped
Load step 0.3 to 2.2 kW at 3 h	AC on within 5 min of the step	OK	3.0 min
	cold side peak below the warning	OK	peak 20.0 degC
Office air 1 x NF-A8	office at least 1.0 K cooler	OK	office 28.6 degC, 1.4 K under its uncooled 30
	rack pulled down for the office, never near its warning	OK	cold side averages 17.1 degC (target 18), max 24.8
	no alarms once settled	OK	none
Office air 2 x P8 Max push-pull	office at least 1.8 K cooler	OK	office 27.6 degC, 2.4 K under its uncooled 30
	rack pulled down for the office, never near its warning	OK	cold side averages 16.9 degC (target 18), max 24.7
	no alarms once settled	OK	none
Office air rack first	office fan stopped whenever the rack ran warm	OK	1.2 h over target + 1.5 K, fan on in 0; off 3.8 h of 5
	cold side peak below the warning	OK	peak 23.7 degC

COMMISSIONING

1 Bench: power the carrier from 12 V with U1 out: +5 V, +12 V on J2. Fit U1: +3V3.
2 Flash both boards over USB-C (esphome run ...); 'Rack link' turns green.
3 Probes: the node log lists every DS18B20 per bus at boot. Warm one at a time in your hand, put each address in sra16-node.yaml, re-flash.
4 AC protocol: point the AC remote at the node box lid: the log says 'Received Coolix' (keep coolix) or 'Midea' (set midea_ir). Re-flash.
5 Press 'Re-send to AC' on the Alarms screen: the AC must go to cool at the setpoint shown. Try OFF and back to AUTO. Stick the emitter where this always works.
6 Doors and float: open / lift each; its dot turns amber. Backwards? add 'inverted: true' to that input.
7 Unplug a probe lead: 'probe missing', 3 beeps a minute, AC kept on.
8 Shutdown: set the limit to 28, warm T1 and T2: K1 closes 120 s later. Only then wire J3 to the UPS / server input. Reset with 'Reset alarms'.
9 Office air: warm T7 in your hand. 'Office air' shows Waiting, then Cooling once the rack is cold; the fan spins up and its rpm shows. Unplug W9: fan alarm.
10 Leave it a day: the cold-side average should sit at the target.

SIMULATED DAYS (renders/controls_sim.png)



Lines: hot side (red), cold side (blue), AC supply and return. Blue band: compressor running; grey: AC off; triangles: IR frames sent; red: alarms. tools/sim_controller.py compiles sra_control.h for the PC and runs it against a lumped model of the rack and the AC - thermostat and restart delay included.

PROJECT	SRA-16 Silent AC Rack - control electronics kit
TITLE	Control logic, alarms, simulation, commissioning
DOC NO.	CP-SRA16-ELC-001 SHEET 5 OF 5 REV B
SCALE	NTS @ A3 UNITS mm LAYOUT v1.3.0
DATE	2026-10-16 DRAWN Claude (AI) for Carbon Project
STATUS	PROTOTYPE - SELV only (12 V); no mains wiring in the kit